A COMPANION NOTE KIRMSE–COXETER AND THE σ-TWIST, BY CONCRETE EXAMPLE

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Abstract. This note accompanies the paper "An order on the Leech lattice from transposition-twisted octonions." Its purpose is purely expository: to walk through the construction step by step, pairing every abstract definition with a concrete hand-checkable example, and to do so first at the E_8 level (where Coxeter's correction of Kirmse provides the historical precedent) and then at the Leech level (where the σ-twist plays the analogous role). No new results are proved. Readers who are comfortable with the main paper will find nothing new here; readers who want to see every piece verified by hand will find the explicit numerics they may have missed.

1. Preliminaries

This note follows the conventions of the main paper. We work with the real octonion algebra $\mathbb{O}$ on $\mathbb{R}^8$ with basis $\{e_0, e_1, \ldots, e_7\}$, and with Wilson's construction of the Leech lattice inside the triple octonion space $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$.

The guiding principle throughout is: *distinguish abstract structure from the specific lattice to which it is applied.* The same ambient algebra can host many lattices; the same lattice can be tested against many bilinear products on the ambient space. "Closure of this lattice under this product" is a question about both, not just one.

2. The E_8 lattice

Abstract. The lattice E_8 is the unique even unimodular lattice in $\mathbb{R}^8$ (up to isomorphism). "Even" means every vector $v \in E_8$ has integer squared norm $\|v\|^2 \in 2\mathbb{Z}$. "Unimodular" means E_8 equals its own dual lattice (the set of vectors having integer inner product with every element of E_8). It contains 240 vectors of minimal squared norm 2, the *roots*.

Concrete. Following Wilson, take for L the integer span of the following 240 octonions:

- **112 "type-1" roots** of the form $\pm e_a \pm e_b$ with $0 \le a < b \le 7$;

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- **128 "type-2" roots** of the form $\frac{1}{2}(\varepsilon_0 e_0 + \varepsilon_1 e_1 + \cdots + \varepsilon_7 e_7)$ where each $\varepsilon_i \in \{+1, -1\}$ and the number of -1s is odd.

Equivalently,

$$L = \{v \in \mathbb{Z}^8 : \sum_i v_i \text{ is even}\} \cup \{v \in (\mathbb{Z} + \tfrac{1}{2})^8 : \sum_i v_i \text{ is odd}\}.$$

Example. *The vector* $(1,1,0,0,0,0,0,0) = e_0 + e_1$ *is a type-1 root: integer coordinates, sum* 2*. The vector* $\frac{1}{2}(-1,1,1,1,1,1,1,1)$ *is a type-2 root: half-integer coordinates, one* -1 *(odd count), coordinate sum* 3 *(odd). Both lie in* L*; both have squared norm* 2*.*

Example. *Distinguished element in* L *to be used throughout:*

$$s \;=\; \tfrac{1}{2}(-e_0 + e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7),$$

of squared norm 2*. In coordinates* $s = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$*: half-integer, one* $-1/2$ *(odd count), sum* 3 *(odd). Confirmed in* L*.*

3. The octonion algebra

Abstract. The octonions $\mathbb{O}$ are the unique real *normed division algebra* of dimension 8: a real 8-dimensional vector space with a bilinear product $x \cdot y$ and a positive-definite quadratic norm satisfying $\|x \cdot y\| = \|x\| \cdot \|y\|$. The algebra is non-commutative and non-associative (but alternative).

Concrete. Fix the basis $\{e_0, e_1, \ldots, e_7\}$ with e_0 the multiplicative identity. The multiplication on the imaginary units is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{1}$$

read cyclically: for (a,b,c) above, $e_a \cdot e_b = +e_c$, $e_b \cdot e_c = +e_a$, $e_c \cdot e_a = +e_b$, with opposite signs for the reversed products. Every imaginary unit squares to $-e_0$.

Example. *From the triple* $(1,2,4)$*:* $e_1 \cdot e_2 = +e_4$ *and* $e_2 \cdot e_1 = -e_4$*. From* $(7,1,3)$*:* $e_1 \cdot e_3 = +e_7$*. From* $(6,7,2)$*:* $e_7 \cdot e_2 = +e_6$*, so* $e_2 \cdot e_7 = -e_6$*.*

Closure of L under the octonion product. Under the Fano convention (1), the lattice L is closed under octonion multiplication: for every $u, v \in L$, the product $u \cdot v \in L$. This is a classical result of Coxeter (1946), verified in the main paper by computing all $8 \times 8 = 64$ products of a $\mathbb{Z}$-basis of L in exact arithmetic and checking that each lies again in L.

Example. *A sample product. From* $s \cdot s$ *with* s *as above, using the identity* $(\sum_{k=1}^{7} e_k)^2 = -7e_0$*:*

$$s \cdot s \;=\; -\tfrac{3}{2}\, e_0 - \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7).$$

Its coordinates are all half-integer; the coordinate sum is $-\frac{3}{2} - 7 \cdot \frac{1}{2} = -5$*, odd. Hence* $s \cdot s \in L$*.*

4. The Kirmse–Coxeter precedent

Historical remark. Kirmse (1924) proposed a specific set of 240 integral octonions together with a multiplication table. He asserted the set was closed under the multiplication. Coxeter (1946) noticed a subtle error: Kirmse's multiplication table, as written, did not produce closure. Coxeter showed that if one permutes two basis indices in the Fano-plane labelling, one obtains a (different) multiplication table under which the 240-element set *is* closed. There are seven such Coxeter-corrected conventions, one for each pair of imaginary indices; each gives one of the seven maximal orders of the integral octonions discussed in Conway–Smith.

The Fano triples (1) used in this note and in the main paper are one of Coxeter's seven corrected conventions. Under this convention L closes, as stated above. We do not reproduce Kirmse's original table here; the point of recalling the story is the structural pattern, not the historical arithmetic:

Observation. *A convention of octonion multiplication fixes how basis products are assigned. A different convention, obtained by transposing two basis indices, assigns products differently. For some lattices, the two conventions are interchangeable (the lattice is closed under both); for others, only one convention closes. Which lattices behave which way is precisely the question that Kirmse got wrong and Coxeter got right.*

A concrete lattice where the story does matter. For the lattice L itself, this distinction is invisible: L is symmetric under every coordinate permutation, so transposing two basis indices produces an algebra under which L also closes. To see the Kirmse–Coxeter phenomenon concretely at the E_8 level we need a lattice that is *not* symmetric under such a swap. One appears naturally in Wilson's Leech construction: the sublattice

$$Ls \;=\; \{\ell \cdot s : \ell \in L\},$$

the image of L under right-multiplication by the element s above. This is again a full-rank sublattice of L, of index 16 (see the main paper, Section 2.3). It is a lattice, so of course it is closed under addition; the question is whether it is closed under octonion multiplication.

Example (Standard product fails on Ls)**.** *Take*

$$a = (-1,\, 0,\, 0,\, 0,\, 1,\, 0,\, 1,\, 1), \qquad b = (-1,\, 1,\, 0,\, 0,\, 0,\, 1,\, 0,\, 1).$$

Both a and b are in Ls (they appear as basis vectors of Ls produced in the main paper's exact-arithmetic verification). Their octonion product, computed under the Fano convention (1)*, is*

$$a \cdot b = (0,\, -2,\, 2,\, 1,\, -2,\, -1,\, -1,\, -1).$$

This vector has integer coordinates and lies in L — but it is not *an integer combination of a $\mathbb{Z}$-basis of Ls. The verification is carried out with exact rational arithmetic in the script* `symbolic_proof_checks.py`*.*

Consequently, $Ls \cdot Ls \not\subseteq Ls$.

This is the structural echo of Kirmse's situation at the sublattice level: a specific sublattice, under a specific octonion product, fails to close.

5. The Coxeter transposition, abstractly

Abstract. Fix a transposition $\sigma = (s\ t)$ on the imaginary basis indices $\{1, \ldots, 7\}$ (for example, $\sigma = (1\ 2)$). Extend σ linearly to a map $\mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and swaps $e_s \leftrightarrow e_t$. This σ is an *isometry* of $\mathbb{R}^8$: it preserves the Euclidean inner product. It is also an involution: $\sigma^2 = \mathrm{id}$.

Starting from the standard product $\cdot$, the map σ induces a *σ-twisted product* on the same $\mathbb{R}^8$:

$$x \cdot_\sigma y \;=\; \sigma\big(\sigma(x) \cdot \sigma(y)\big). \tag{2}$$

The pair $(\mathbb{R}^8, \cdot_\sigma)$ is again an octonion algebra, abstractly isomorphic to $(\mathbb{R}^8, \cdot)$ via σ itself. But it is a *different bilinear product* on the same $\mathbb{R}^8$: it assigns different specific products to pairs of basis vectors.

Concrete. Take $\sigma = (1\ 2)$, which swaps e_1 and e_2 (and fixes $e_0, e_3, e_4, e_5, e_6, e_7$). Then

$$e_1 \cdot_\sigma e_2 = \sigma(\sigma(e_1) \cdot \sigma(e_2)) = \sigma(e_2 \cdot e_1) = \sigma(-e_4) = -e_4,$$

whereas $e_1 \cdot e_2 = +e_4$ under the standard product. Similarly one checks $e_1 \cdot_\sigma e_4 = +e_2$ (vs. $-e_2$ under $\cdot$), and so on: the two products disagree on some pairs of basis elements.

What is twisted and what is invariant.

- **Twisted:** the multiplication table. Under $\cdot_\sigma$, the Fano triples (1) are replaced by $(\sigma(a), \sigma(b), \sigma(c))$ for each original triple.
- **Invariant under σ (as a linear map):** the underlying vector space $\mathbb{R}^8$, the Euclidean inner product, the decomposition into integer and half-integer vectors, and the lattice L itself ($\sigma(L) = L$ since σ is a coordinate permutation).
- **Not invariant under σ:** specific sublattices of L that depend on a distinguished element, such as Ls. Since Ls is defined as $\{\ell \cdot s : \ell \in L\}$, applying σ gives

  $$\sigma(Ls) = \{\sigma(\ell) \cdot_\sigma \sigma(s) : \ell \in L\} \;\neq\; Ls$$

  in general. The main paper verifies $\sigma(Ls) \neq Ls$ directly: for $\sigma = (1\ 2)$, the vector $(-1, 0, 1, 0, 0, 1, 0, 1)$ lies in $\sigma(Ls)$ but not in Ls.

The last two bullets together say: *σ is an isometry of $\mathbb{R}^8$ that fixes L but moves Ls to a distinct sublattice.*

6. The twist at the E_8 level changes the story

The sublattice $\sigma(Ls)$ is the image of Ls under the transposition. It has the same rank as Ls, the same index in L, and is isometric to Ls. But its elements are different vectors. The question of closure under the standard octonion product is now posed afresh.

Example (Standard product succeeds on $\sigma(Ls)$)**.** *A $\mathbb{Z}$-basis of $\sigma(Ls)$ is obtained by applying σ (coordinate swap of positions $1, 2$) to each basis vector of Ls. Using the basis generated in the main paper's verification, all $8 \times 8 = 64$ products of basis vectors of $\sigma(Ls)$, under the standard octonion product $\cdot$, lie again in $\sigma(Ls)$. The verification is in* `symbolic_proof_checks.py` *as Lemma D.*

Hence $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.

Two equivalent readings. Because σ is an isometry and an algebra isomorphism $(\mathbb{R}^8, \cdot) \to (\mathbb{R}^8, \cdot_\sigma)$, one can restate Example 6 in an equivalent form on the original lattice Ls with the twisted product:

$$Ls \cdot_\sigma Ls \;\subseteq\; Ls.$$

(Apply σ to both sides of the inclusion $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ and use $\sigma^2 = \mathrm{id}$ together with (2).) The two forms are tautologically equivalent.

Reading both against Example 4:

$$\underbrace{Ls \cdot Ls \not\subseteq Ls}_{\text{standard product fails on } Ls} \qquad \text{versus} \qquad \underbrace{Ls \cdot_\sigma Ls \subseteq Ls}_{\text{twisted product succeeds on } Ls} \quad .$$

Equivalently:

$$\underbrace{Ls \cdot Ls \not\subseteq Ls}_{Ls \text{ fails}} \qquad \text{versus} \qquad \underbrace{\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)}_{\sigma(Ls) \text{ succeeds}} .$$

Interpretation. Two different bilinear products on $\mathbb{R}^8$ (namely $\cdot$ and $\cdot_\sigma$) are isomorphic as abstract algebras. Every theorem phrased in the language of the abstract algebra carries over. But "does lattice X close under product $\star$?" is not a theorem phrased purely in the abstract algebra: it also names a specific subset $X \subset \mathbb{R}^8$, and the answer depends on how the isomorphism relates X to itself.

If $\sigma(X) = X$, the answer is the same for both products. If $\sigma(X) \neq X$, the answer can flip: X may fail to close under $\cdot$ while $\sigma(X)$ closes under $\cdot$; equivalently, X closes under $\cdot_\sigma$ while $\sigma(X)$ fails under $\cdot_\sigma$. The lattice L itself is σ-invariant and therefore closes under both products. The sublattice Ls is *not* σ-invariant, and this is exactly what opens the gap between the two closure statements.

This is the structural lesson of Kirmse–Coxeter in isolation. The same lesson, applied one dimension higher, is the content of the main paper.

7. Wilson's Leech lattice

Abstract. The Leech lattice Λ is the unique even unimodular lattice of rank 24 with no roots (i.e. no vectors of squared norm 2). Its minimal vectors have squared norm 4, and there are exactly 196,560 of them.

Concrete (Wilson's construction). Identify $\mathbb{R}^{24}$ with $\mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$: three copies of the octonions, labelled $\alpha = 1, 2, 3$. Let $\bar{s} = \frac{1}{2}(-e_0 - e_1 - e_2 - \cdots - e_7)$ (the octonion conjugate of the negative of the earlier s; it also lies in L). Define two sublattices of L:

$$Ls = \{\ell \cdot s : \ell \in L\}, \qquad L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\},$$

each of index 16 in L. Wilson's Leech lattice is then

$$\Lambda \;=\; \big\{(x, y, z) \in L^3 \;:\; x + y,\; x + z,\; y + z \;\in\; L\bar{s};\quad x + y + z \;\in\; Ls\big\}. \tag{3}$$

The sublattice Ls from Section 4 is precisely the "condition-(3) block sum" lattice in Wilson's definition.

8. The standard triple product on $\mathbb{R}^{24}$

Abstract. The natural bilinear product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ built from a single octonion product is a *triple product with $\mathbb{Z}_3$-symmetric cross-block routing.* For $u = (x, y, z)$ and $v = (x', y', z')$ define

$$u \star v \;=\; (P_1, P_2, P_3),$$

where

$$\begin{aligned} P_1 &= x \cdot x' + z \cdot y' + y \cdot z', \\ P_2 &= y \cdot y' + x \cdot z' + z \cdot x', \\ P_3 &= z \cdot z' + y \cdot x' + x \cdot y'. \end{aligned}$$

Same-block products stay in the block; cross-block products $O_\alpha \times O_\beta$ land in O_γ with $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$. The three blocks play the same role under cyclic rotation $1 \to 2 \to 3 \to 1$.

Why condition (3) is the obstruction. A direct computation from the definitions gives

$$P_1 + P_2 + P_3 \;=\; (x + y + z) \cdot (x' + y' + z').$$

Thus "$u \star v$ satisfies Wilson's condition (3)" amounts to: "the product of two elements of Ls lies in Ls".

Using $u, v \in \Lambda$ means $x + y + z \in Ls$ and $x' + y' + z' \in Ls$, so the question reduces exactly to whether $Ls \cdot Ls \subseteq Ls$. By Example 4, it does not. Hence:

Example (Standard triple product fails on Λ)**.** *There exist $u, v \in \Lambda$ for which $u \star v \notin \Lambda$: the block sum of $u \star v$ need not lie in Ls, failing Wilson's condition (3). The concrete obstruction is precisely the pair (a, b) of Example 4: any $u, v \in \Lambda$ whose triple sums equal a, b respectively produce a $u \star v$ whose block sum equals $a \cdot b \notin Ls$.*

9. The σ-twisted triple product

Abstract. Apply σ componentwise to each of the three octonion blocks. This yields a linear map on $\mathbb{R}^{24}$, still denoted σ, that is an isometry and satisfies $\sigma^2 = \mathrm{id}$. Replace the octonion product in each of the nine block-pair products of $\star$ by the σ-twisted product $\cdot_\sigma$ from (2): this produces a new bilinear product $\star_\sigma$ on $\mathbb{R}^{24}$. By the same formal manipulation as before,

$$u \star_\sigma v \;=\; \sigma\big(\sigma(u) \star \sigma(v)\big),$$

where on the right $\star$ is the standard (untwisted) triple product.

Concrete. Choose $\sigma = (1\ 2)$, acting on imaginary indices of each octonion block. Then for $u, v \in \Lambda$,

$$(\text{block sum of } u \star_\sigma v) \;=\; \sigma\big((\sigma(x) + \sigma(y) + \sigma(z)) \cdot (\sigma(x') + \sigma(y') + \sigma(z'))\big).$$

Inside the outer σ, each triple sum lies in $\sigma(Ls)$ (since each original triple sum lies in Ls, by Wilson's condition (3) on u, v). By Example 6, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$. Applying the outer σ brings the result into $\sigma(\sigma(Ls)) = Ls$.

So the block sum of $u \star_\sigma v$ lies in Ls: Wilson's condition (3) is restored. Conditions (1) and (2) go through by analogous arguments (see the main paper, Section 4). The net effect is:

Example (σ-twisted triple product succeeds on Λ)**.** *For all $u, v \in \Lambda$, the product $u \star_\sigma v$ lies in Λ. This is Theorem 3.1 of the main paper.*

What is twisted and what is invariant (at the Leech level).

- **Twisted:** the octonion multiplication used inside each of the nine block-pair products. Equivalently, the Fano triples (1) are replaced block by block by their σ-images.
- **Invariant under σ:** the ambient space $\mathbb{R}^{24}$, the three-block decomposition into $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$, the $\mathbb{Z}_3$ cross-block routing rule, and the sub-lattice L of each block ($\sigma(L) = L$).
- **Not invariant under σ:** the sub-lattice Ls of each block ($\sigma(Ls) \neq Ls$), and through it the Leech lattice itself ($\sigma(\Lambda) \neq \Lambda$ in general, since Λ is defined by a condition involving Ls).

The comparison with the E_8 level is exact. One level down, σ fixes L but not Ls, and this is what allows the twist to change the closure question for Ls. One level up, σ fixes $\mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ but not Λ, and this is what allows the twist to change the closure question for Λ.

10. Summary of concrete data

Three empirical facts are verified by exact rational arithmetic in `symbolic_proof_checks.py`:

(1) $\sigma(Ls) \neq Ls$ (explicit witness: $(-1, 0, 1, 0, 0, 1, 0, 1) \in \sigma(Ls) \setminus Ls$).
(2) $Ls \cdot Ls \not\subseteq Ls$ under the standard octonion product (explicit witness a, b in Example 4 with $a \cdot b \notin Ls$).
(3) $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ under the standard octonion product (all 64 basis products land in $\sigma(Ls)$).

Two equivalent phrasings of the main result are:

- "Λ is closed under the σ-twisted triple product $\star_\sigma$."
- "$\sigma(\Lambda)$ is closed under the standard triple product $\star$."

Both say the same thing via the algebra isomorphism $\sigma : (\mathbb{R}^{24}, \star) \to (\mathbb{R}^{24}, \star_\sigma)$; neither says that Λ is closed under $\star$, which is false.